

Turbulent Times: Uncovering the Origins of US Natural Gas Price Fluctuations Since 1993

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Abstract

In this paper, we investigate supply and demand shocks in the US natural gas market, focusing on how the effects of these shocks have changed over time. Using a sign-identified structural vector autoregression (SVAR) model that allows for both time-varying parameters and stochastic volatility, we decompose the real price of natural gas into supply shocks, aggregate demand shocks driven by changes in the US economic activity, precautionary inventory demand in anticipation of changes in future demand-and-supply conditions, and residual demand shocks not otherwise accounted for by the previous three shocks. Using quarterly data from 1976 to 2015, we find that an unanticipated supply disruption raises natural gas prices, reduces the aggregate economic demand, and lowers the precautionary inventory demand, while negative aggregate demand shocks, on the other hand, depress natural gas prices, reduce natural gas production, and increase precautionary inventory demand. We also find that following a negative precautionary inventory demand shock, aggregate demand driven by real economic activity declines marginally, and the marketed natural gas production and real prices decrease as well. Our results further suggest that such impact responses have evolved considerably over time with changing market conditions.

Keywords: natural gas, time-varying, stochastic volatility, supply disruptions, demand shocks, inventory, price movements, structural vector autoregressions

JEL codes: E31, E37, Q41, Q43

“We produce more natural gas than ever before -- and nearly everyone’s energy bill is lower because of it. And over the last four years, our emissions of the dangerous carbon pollution that threatens our planet have actually fallen.”

President Barack Obama, State of the Unions Speech, February 12, 2013

1. Introduction

Natural gas is a critical component of the energy industry in the US. The Energy Information Administration (EIA) estimated that in 2014, approximately 26.7 trillion cubic feet of natural gas was consumed in the US, supplying roughly 27% of electricity generation and 19% of all residential heating¹. In commercial manufacturing, natural gas is widely used as both a primary fuel source and raw material. At the macro-scale, the domestic natural gas industry is closely tied to economic development, employment, and industrial output, and its growth is often considered as contributing to US energy independence. In his 2013 State of the Unions speech, President Obama claimed that the domestic production of natural gas “has led to cleaner power and greater energy independence”. As the burning of natural gas emits fewer greenhouse gas emissions relative to coal, the following year he highlighted natural gas as the “bridge” fuel that can transition the US economy from its traditional reliance on fossil fuels.

However, little is known about the price fluctuations in the US natural gas market and their specific causes. As figure 1 demonstrates, natural gas prices in the US have experienced significant volatility over the past three decades. Behind these price fluctuations are fundamental structural changes in the natural gas market driven mainly by exogenous supply and demand shocks. In the aftermath of the supply disruptions caused by hurricanes Katrina and Rita in 2005, natural gas prices hit a record high in both nominal and real terms, followed by a more than 50% drop as weather conditions unexpectedly improved². Then during the financial crisis of 2008, natural gas prices experienced significant shifts, a pattern shared by many other commodities. Most recently, domestic production from shale and other lower-permeability formations has become economically profitable, sparked by the popularization of the combined use of horizontal drilling and hydraulic fracturing techniques. These new techniques have completely revolutionized the market. In 2007 shale production roughly 10% of US dry natural gas

¹ See EIA natural gas consumption data at: https://www.eia.gov/dnav/ng/ng_cons_sum_dcunusa.htm

² See EIA price data at: <https://www.eia.gov/dnav/ng/hist/n9190us3m.htm>

production, while in 2014 it increased to over 53%³. Despite these enormous changes, studies of the driving forces behind natural gas price movements still lag behind those from other energy markets.

The purpose of this paper is to disentangle supply and demand shocks in the US natural gas market and investigate how the effects of these shocks have changed over time. Using a sign-identified structural vector autoregression (SVAR) model that allows for both time-varying parameters and stochastic volatility (Baumeister and Peersman, 2013a, b), we decompose the real price of natural gas into supply shocks, aggregate demand shocks driven by changes in the US economic activity, precautionary inventory demand shocks in anticipation of future demand-and-supply conditions, and residual demand shocks not otherwise accounted for by the previous three shocks. Using quarterly data from 1976 to 2015, we estimate the time-varying effects of these shocks on the evolution of US natural gas prices, calculate price elasticities due to shifts in the supply or demand curve associated with each structural shock, and evaluate their relative importance on price variations during historical episodes characterized by large price fluctuations.

Overall, estimation results suggest that the impacts of different structural shocks have evolved considerably since markets were formally deregulated in 1993. We find that an unanticipated supply disruption raises natural gas prices, reduces the aggregate economic demand, and lowers precautionary inventory demand. Negative aggregate demand shocks, on the other hand, depress natural gas prices, reduce natural gas production, and encourage precautionary inventory demand. Additionally, we find that following a negative precautionary inventory demand shock, the marketed natural gas production and real prices declines significantly, while aggregate demand only declines marginally. Regarding the time variation in the impact response, we find that both the price elasticities of supply and demand have increased in magnitudes in recent years, possibly reflecting the greater flexibility in production and in fuel switching due to technological advances. Though precautionary inventory demand is shown to have a greater impact over time, its magnitude remains small. The main drivers of natural gas price movements appear to be the supply and aggregate demand shocks.

The remainder of this paper is organized as follows. The next section briefly reviews relevant literature on estimating the supply and demand structure in the natural gas market. Section three describes the econometric procedure used in the analysis, and section four presents the data used in this analysis. In section five, we report the estimation results from a time-varying structural VAR model that allows for stochastic volatility. In section six, we investigate a few historical episodes and estimate the relative role of supply and demand in these events, and section six concludes the paper.

³ See shale gas production data at: https://www.eia.gov/dnav/ng/ng_prod_shalegas_s1_a.htm.

2. Brief Literature Review

Only a handful of studies have attempted to understand the supply-and-demand structure in the US natural gas market, with the majority focusing on the spatial pricing efficiency in regional gas markets and natural gas' price relationship with other energy commodities. For instance, Mohammadi (2011) demonstrates that while prices of oil and coal are determined globally and by long term contracts respectively, natural gas prices are determined regionally. Olsen et al. (2015) find cointegration between North American natural gas markets, though the degree of integration varies depending on their locations. Siliverstovs et al. (2005) find integration of natural gas prices within European and US markets, but not between them. Similar results are found in Renou-Maissant (2012) and Park et al. (2007), among others. The pricing relationship between natural gas and crude oil and other energy commodities has as well received much attention from the literature, as for many applications, natural gas and refined petroleum products are close substitutes. Though fuel switching may be limited in the short-run due to technological constraints, researchers overall agree that (1) natural gas and crude oil prices are strongly linked before 2008, and (2) natural gas prices tend to be more influenced by oil prices, rather than the other way around (e.g. Atil et al. (2014), Brigida (2014), Brown and Yücel (2008), Hartley and Medlock (2014), Ji et al. (2014), and Ramberg and Parsons (2012)).

A core problem in the natural gas market is understanding how natural gas prices respond to different market shocks and the relative importance of each shock in driving price movements. Recently, Nick and Thoenes (2014) analyze the German natural gas market, and the impact that three significant supply shocks have had on prices using a structural VAR model. They argue that in the short-run temperature, storage, and supply shortfalls play an important role. Woo et al. (2014) find that end-use prices of natural gas generally reflect cost of wholesale, with only a small time-period of adjustment. Mu (2007) highlights the importance of weather on natural gas prices, as demand is influenced both directly by cold weather for heating as well as indirectly by warm weather through increased electricity demand. Hulshof et al. (2016) conclude that the day-ahead gas prices at the Dutch gas hub are predominantly determined by gas-market fundamentals. Brown and Yücel (2008), by contrast, argue that after accounting for weather, inventories, and supply disruptions, oil price fluctuations remain the primary determinants of natural gas prices.

Given that the market for natural gas has undergone significant changes over the past decades, studies have also attempted to include structural breaks when modeling natural gas prices. Qin et al. (2010) find that the importance of fundamentals change depending on market regime, playing a larger role in bullish than in bearish markets. They further contend that natural gas price behavior is far more complicated than that predicted by fundamentals, and that volatility unexplained by fundamentals plays

an essential role in natural gas price behavior. Several studies have also incorporated the impact of shale boom in shaping natural gas prices. Arora (2014) uses a structural VAR to estimate the supply and demand elasticity of US natural gas prices prior to and after the shale boom. He finds that both the short- and long-run natural gas supply becomes more elastic when the effects of a shale development are included, while the demand becomes either less responsive or similar to price changes after accounting for shale production (post-2007). Wakamatsu and Aruga (2013) model the US and Japanese natural gas markets using a one-time structural break from the shale boom, finding that the two markets used to be interlinked before the revolution, but the US market has become more independent after the revolution.⁴

Our analysis is closely related to this last stream of literature, focusing on the time-variation and structural breaks in the natural gas supply and demand dynamics. This work is of particular importance given the increasingly volatile natural gas market that has witnessed significant changes in the market structure over the past two decades. The studies of Wakamatsu and Aruga (2013) and Arora (2014) both impose a one-time structural break to account for the shale gas boom. However, a more comprehensive structure is needed to account for other exogenous shocks that have significantly influenced the natural gas markets. Additionally, as suggested by Primiceri (2005), imposing discrete structural breaks may be inappropriate as aggregation in the private sector can smooth such changes.

Our work contributes to the growing literature on the natural gas markets in a number of ways. First, we construct a comprehensive framework that enables a thorough investigation of the role of supply disruptions, aggregate demand shocks, and speculative inventory shocks on natural gas prices in the US. Second, recognizing both that massive changes have occurred in the natural gas market and that these changes are likely to evolve gradually over time, we consider a time-varying structural VAR model that allows for smooth, continuously-evolving parameters (Baumeister and Peersman, 2013a, b). The estimation procedure enables us to estimate the dynamic effects of various structural shocks on the evolution of natural gas prices. Third, given the apparent changes in natural gas price volatility (figure 1), we explicitly consider residual heteroscedasticity in the VAR model that accounts for both changes in the magnitude of structural shocks and their immediate impacts. Such heteroscedasticity, if overlooked, may generate spurious inference from the estimated coefficients (Cogley and Sargent, 2005). Finally, unlike previous structural VAR models that often rely on a contemporaneous exclusion or recursive identification strategy, we use sign-restrictions directly implied from economic theory to recover structural shocks. The resulting impulse responses, forecast error variance decompositions, historical

⁴ In a slightly different vein, Arora and Lieskovsky (2014) find that the shale gas revolution has made the natural gas industry more relevant to the US economic activity by estimating a structural VAR model accounting for different timeframes.

decompositions, and elasticity estimates complement the existing literature on modeling the supply and demand structure in the US natural gas market.

3. Methodology

Consider the following reduced-form vector autoregression (VAR) model with a lag length of p :

$$(1) \quad y_t = B_{1,t} y_{t-1} + \dots + B_{p,t} y_{t-p} + e_t = X_t' \theta_t + e_t,$$

where y_t is a $[4 \times 1]$ vector of endogenous variables consisting of the total physical availability of natural gas in the US ($prod_t$), aggregate real economic activity (rea_t), the precautionary inventory demand (inv_t), and the real price of natural gas (rpo_t). The right-hand side of equation (1) can be simplified as $X_t' \theta_t + \varepsilon_t$, where θ_t consists coefficients on the lags of the endogenous variables $B_{1,t}, \dots, B_{p,t}$, each of which is a $[4 \times 4]$ matrix. The reduced-form residuals e_t can be written as weighted averages of the structural errors u_t , which are commonly assumed to be independently and identically distributed (*iid*). Following Kilian (2009); Kilian and Murphy (2012); Arora (2014); Arora and Lieskovsky (2014); Kilian and Murphy (2014), we define our four structural shocks as shocks in supply, aggregate demand, speculative inventory demand, and residual demand. The relationship between reduced-form and structural innovations may be written as

$$(2) \quad e_t \equiv \begin{pmatrix} e_t^{prod} \\ e_t^{rea} \\ e_t^{inv} \\ e_t^{rpo} \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{pmatrix} u_t^{supply\ shock} \\ u_t^{aggregate\ demand\ shock} \\ u_t^{speculative\ inventory\ demand\ shock} \\ u_t^{residual\ demand\ shock} \end{pmatrix},$$

where $a_{11}, a_{12}, \dots, a_{44}$ refer to contemporaneous relations between endogenous variables.

A supply shock refers to unexpected supply disruptions in the natural gas market that displace the supply curve due to technology, weather or policy changes. Examples of recent supply shocks include the loss of offshore production in 2005 due to hurricane Katrina, the rapid increase of production from shale deposits. An economic activity shock incorporates unexpected demand shocks driven by changes in real economic activity in the US not already reflected in supply shocks. Though Kilian (2009) and a number of following papers use an aggregate economic index based on dry cargo shipping rates to represent oil demand shocks driven by global real economic activity, we prefer a regional economic activity measure for the natural gas market. Previous studies have found that unlike oil prices which are determined in the global market, natural gas is a regional commodity influenced primarily by region-specific demand and

supply conditions (e.g. Mohammadi, 2011). The third shock, which we refer to as speculative inventory demand shock, includes shifts in the demand for natural gas inventories not otherwise explained by supply or aggregate demand shocks. If market participants are forward-looking and rational, then speculative inventory demand increases if net prices of natural gas (excluding cost of carry) are expected to increase. Kilian and Murphy (2014) argue that the speculative demand shock reflects changes in inventory behavior due to rational market participants reacting to news about future supply or demand conditions. It should be noted that our definition of speculative activity is rather broad – it includes any inventory accumulating behavior in anticipation of future supply shortfalls relative to market demand. Finally, the residual shock refers to innovations in natural gas demand not otherwise accounted for by the three previous shocks. Examples of such residual demand shocks include changes in inventory technology or preferences, residential consumers’ preferences changes, changes in consumption pattern due to population shift toward warmer regions, etc. Similar VAR specifications have been used in various previous studies to disentangle structural shocks in the crude oil and natural gas markets, including Arora (2014), Arora and Lieskovsky (2014), Baumeister and Peersman (2013a), Kilian (2009), and Kilian and Murphy (2014) among others.⁵

While equation (1) can be consistently estimated using standard econometric procedures, restrictions need to be placed on the contemporaneous coefficient matrix (equation 2) to recover the structural innovations u_t . Previous studies using a structural VAR approach have often imposed a recursive structure of the endogenous variables in the system in which the preceding variable affects the following variable at contemporaneous time, but not *vice-versa*. Implicit in this approach is the assumption that the very short-run aggregate demand or supply elasticity equals zero so that the correlations between the error in forecasting quantity and price changes in the next period could be interpreted as the response of aggregate supply or demand to the structural shock (Hamilton, 2015)

Four problems exist with such recursively-identified SVAR models. First, previous studies have often assumed that parameters in an SVAR model remain constant throughout the sample period. As argued earlier, this assumption is unlikely to hold due to the dynamic nature of the natural gas market, especially since 1993. Second, even for studies that allow for time-varying parameters, the conventional approach has been to impose a one-time or multiple abrupt structural breaks in the data. While such

⁵ Previous studies on natural gas market also consider the impact of weather conditions on price fluctuations. Since our model is estimated on a quarterly basis, short-term weather impacts are likely to be small. In the empirical part, we also deseasonalize our data using quarterly dummies, further removing the seasonality in demand and inventory due to weather variations. Finally, even if weather does play a significant role in the deseasonalized quarterly data, it should show up in the residual demand shocks from residential heating demand.

discrete break models may describe the occurrence of unexpected, rapid policy changes in the market, rarely do commodity markets evolve in an abrupt fashion such that the impacts of structural shocks differ dramatically prior to and after the break. In particular, the discrete break model cannot capture changes brought about by the private sector where heterogeneous agents respond to external shocks in a diverse fashion. Aggregation of these heterogeneous behaviors are likely to smooth out the economic impacts of discrete policy changes (Primiceri, 2005). Additionally, if economic agents are rational and forward-looking, the expected policy changes are likely to be incorporated into business forecasting long before the change actually takes place. The learning dynamics of the private sector imply that models only allow for discrete structural breaks are unable to capture how modifications occur in the market transmission mechanism.

A third problem with previous studies is that the error terms in the VAR models are often assumed to be identically distributed (homoskedastic) over time, an assumption that rarely holds in practice. As figure 1 illustrates, natural gas prices have undergone considerable volatility over the past decades. Such heteroscedasticity in the underlying data, if overlooked, may generate spurious dynamics in the coefficients of the model (Cogley and Sargent, 2005). Finally, in a recursively-identified SVAR, structural shocks (u_t) are recovered by orthogonalizing the reduced-form errors (e_t) through a Cholesky decomposition. This approach relies on the assumption that the relationship between endogenous variables follows a particular causal chain such that the preceding variable affects the following variable at contemporaneous time, but not the other way around. This assumption is less appropriate when dealing with data of low frequency (e.g. quarterly). Additionally, unless there is a convincing theoretical rationale for a particular ordering, the resulting analyses for the structural VAR model (e.g. impulse responses, variance decompositions, etc.) are not credible. This problem cannot be alleviated by considering alternative orderings, as restrictions on structural parameters should follow economic rationales rather than sensitivity analysis (Kilian, 2013).

3.1 A Time-Varying Parameter VAR Model with Stochastic Volatility

Baumeister and Peersman (2013a, 2013b) suggest a VAR model with time-varying parameters and stochastic volatility to resolve the first three problems. Specifically, instead of assuming constant parameters or abrupt discrete changes in parameters over the sample period, they let the coefficient matrices $B_{1,t}, \dots, B_{p,t}$ in equation (1) to vary in time, or equivalently, θ_t to be time-varying. Further, the disturbance term e_t is assumed to be normally distributed with mean zero and a time-varying covariance matrix Ω_t :

$$(3) \quad \Omega_t = A_t^{-1} H_t (A_t^{-1})',$$

where A_t is a 4×4 lower triangular matrix that models the contemporaneous correlations among the four endogenous variables and H_t is a 4×4 diagonal matrix that models the stochastic volatility in the residuals. The specific representations of A_t and H_t are shown in equation (4):

$$(4) \quad A_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \alpha_{21,t} & 1 & 0 & 0 \\ \alpha_{31,t} & \alpha_{32,t} & 1 & 0 \\ \alpha_{41,t} & \alpha_{42,t} & \alpha_{43,t} & 1 \end{bmatrix} \quad H_t = \begin{bmatrix} h_{1,t} & 0 & 0 & 0 \\ 0 & h_{2,t} & 0 & 0 \\ 0 & 0 & h_{3,t} & 0 \\ 0 & 0 & 0 & h_{4,t} \end{bmatrix}.$$

Now let $\alpha_t = [\alpha_{21,t}, \alpha_{31,t}, \alpha_{32,t}, \alpha_{41,t}, \alpha_{42,t}, \alpha_{43,t}]'$ be a vector of elements from A_t that are both nonzero and non-unity; and $h_t = [h_{1,t}, h_{2,t}, h_{3,t}, h_{4,t}]'$ be the diagonal elements of H_t . To impose time variations in the model, they assume θ_t and the free elements from A_t (non-zero and non-unit) to evolve as random walks without drift, and that each element of the vector of volatility h_t to follow a geometric random walk (equations (5)-(6)):

$$(5) \quad \theta_t = \theta_{t-1} + v_t, \quad v_t \sim N(0, Q),$$

$$(6) \quad \alpha_t = \alpha_{t-1} + \zeta_t, \quad \zeta_t \sim N(0, S),$$

$$(7) \quad \ln h_{i,t} = \ln h_{i,t-1} + \sigma_i \eta_{i,t}, \quad \eta_t \sim N(0, 1),$$

where the error terms v_t , ζ_t , and η_t are independent of each other and normally distributed. Further, a block-diagonal structure is imposed for S (the variance–covariance matrix of the error term ζ_t), as in equation (8):

$$(8) \quad S \equiv \text{Var}(\zeta_t) = \text{Var} \left(\begin{bmatrix} \zeta_{21,t} \\ \zeta_{31,t} \\ \zeta_{32,t} \\ \zeta_{41,t} \\ \zeta_{42,t} \\ \zeta_{43,t} \end{bmatrix} \right) = \begin{bmatrix} S_1 & 0_{1 \times 2} & 0_{1 \times 3} \\ 0_{2 \times 1} & S_2 & 0_{2 \times 3} \\ 0_{3 \times 1} & 0_{3 \times 2} & S_3 \end{bmatrix},$$

where $S_1 \equiv \text{Var}(\zeta_{21,t})$, $S_2 \equiv \text{Var}([\zeta_{31,t}, \zeta_{32,t}]')$, and $S_3 \equiv \text{Var}([\zeta_{41,t}, \zeta_{42,t}, \zeta_{43,t}]')$. Since A_t models the contemporaneous correlations among the endogenous variables, the block-diagonal structure of S essentially implies that shocks to the contemporaneous correlations are correlated within, but not across, equations. Compared to the general case of S being unrestricted, the block-diagonal structure has

the advantage of simplifying the inference and increasing the efficiency of the estimation algorithm (Primiceri, 2005).

The specification outlined above allows for time variation in both coefficient estimates and the variance-covariance matrix of the residuals. The former feature allows us to capture possible non-linearities and time variation in the lag structure of the endogenous variables, while the latter enables us to model not only the heteroscedasticity in the residuals but also changes in the contemporaneous relationships among the endogenous variables. These two features combined allow the data to determine whether the time variation in the structural relationship among endogenous variables is due to changes in the size of the shock and its associated contemporaneous impact, or from changes in the shock transmission mechanism.

3.2 Identification of Structural Shocks

Unless the recursive causal chain imposed on endogenous variables is economically justified, a recursively-identified structural VAR model cannot be used to generate plausible economic interpretations (e.g., Kilian, 2013). As an alternative in recent years, a growing number of papers have used sign restrictions for a recursive identification strategy. In a sign-identified structural VAR, each identifiable structural shock is required to generate responses among endogenous variables that follow a unique sign pattern motivated by direct economic theories. Unlike recursive VARs that are exactly identified, a large number of possible solutions are admissible in a sign-identified model as long as the responses of endogenous variables fall within the pre-specified sign pattern. Recent examples of using sign-identified structural VARs to disentangle demand and supply shocks in energy markets include Arora (2014), Baumeister and Peersman (2013a, 2013b), Kilian and Murphy (2012, 2014), among others.

Table 1 summarizes the sign restrictions imposed in the present paper to identify structural shocks in the US natural gas market. Before delving into specific sign restrictions, it is worthwhile to elaborate the inventory variable used in our model. Like many storable commodities, the storable nature of natural gas forces inventories to play a pivotal role in the market dynamics. Along with mitigating the seasonal variation in both production and consumption, the level of inventory available partly reflects how quickly firms can respond to unexpected demand or supply shocks. The theory of storage (e.g., Kaldor, 1939; Working, 1948, 1949; Brennan, 1958) states that firms earn a convenience yield by holding inventory at hand, which prevents disruptions in the flow of goods and services, and in turn reduces production uncertainty. A number of studies have investigated the relationship between prices and inventories (e.g., Wright and Williams, 1991; Miranda and Fackler, 2004), finding that inventories not only help absorb

price fluctuations, but also reflect the degree to which forward-looking inventory holders expect price to rise or fall in a future period. Accordingly, the demand for inventory in the natural gas market may be interpreted a speculative demand, as speculators have the ability to put more (less) into storage in anticipation of a substantial rise (fall) in future prices.⁶

Our assumptions of how short-run demand and supply contemporaneously respond to structural shocks (table 1) follow Arora (2014) and Kilian and Murphy (2014) closely. Conditional on all past data, a positive supply shock shifts the natural gas supply curve to the right along the demand curve, and hence lowers natural gas prices and increases real economic activity within the same time horizon. The responses of natural gas prices and aggregate demand to positive supply shocks at contemporaneous time are thus restricted to be non-positive and non-negative, respectively. Kilian and Murphy (2014) argue that the impact of supply shocks on speculative inventory demand are ambiguous *ex ante*. In the event of supply disruptions, storage holders may sell out inventories as the opportunity cost associated with storing natural gas is high. However, additional natural gas may be put into storage in anticipation of increases in future prices due to supply shortfalls. It is hard to anticipate which effect will dominate the inventory market when supply shocks occur, so no restrictions are placed on the responses of precautionary inventory demand to supply innovations.

Our second structural shock stems from unanticipated innovations to aggregate demand not already captured by supply shocks. Conditional on all past information, a positive aggregate economic demand shock shifts the downward-sloping demand curve to the right along the supply curve. This shock is assumed to not only raise the total supply and real price of natural gas, but also stimulate aggregate real economic activity demand at the same period. Similar to supply shocks, no sign restrictions are placed on the contemporaneous response of speculative inventory demand to aggregate economic shocks, as inventory holders may either increase or decrease their holdings of natural gas.

Our third structural shock is associated with news about unanticipated changes in future supply and demand of natural gas not already embedded in current supply and aggregate economic shocks, which we term “speculative inventory demand shock”. If a tighter future supply-and-demand relationship is anticipated, such a positive speculative demand shock not only increases the inventory demand in the current period, but also shifts the short-run contemporaneous *total* demand curve to the right along the supply curve (conditional on all past information), effectively raising current-period prices. Examples of

⁶ It should be noted that the amount of natural gas in storage reflects the inventory holders’ view toward expected supply shortfalls relative to demand in a future period, rather than future demand or future supply alone.

positive speculative demand shock include the news that (1) recoverable reserves of underground natural gas are less than the numbers released by the EIA, (2) anticipated technical advances that allow natural gas to be used more efficiently and more cleanly, or (3) expectation of increased natural gas exports due to future investment in infrastructure that expands the existing pipeline capacity, to name a few. As inventory accumulates in anticipation of future supply shortfalls relative to demand, current economic activity would be depressed, as less natural gas is available for industrial consumption. Further, the expectation that future supply-and-demand conditions will tighten would encourage larger natural gas production in the present period, facilitating inventory accumulation.

Our last shock is considered a residual demand shock, reflecting unanticipated changes in the demand of natural gas not already captured by supply, aggregate economic activity, and speculative demand shock. As pointed out by Kilian and Murphy (2014), the residual shock is usually hard to interpret economically as it is a conglomerate of idiosyncratic demand shocks in the market. Hence, we do not place any restrictions on the responses of endogenous variables to the residual shock.

3.3 Implementation of the Estimation and Identification Procedure

Following Baumeister and Peersman (2013a, 2013b), the VAR model with time-varying parameters and stochastic volatility is estimated using the Bayesian methods of Kim and Nelson (1999). Below, we briefly describe the estimation procedure.

Setting prior distributions. We first obtain the time-invariant OLS estimates of the VAR model (equation (1)) for a pre-specified training sample. Denote $\hat{\theta}_{OLS}$, $\hat{V}(\hat{\theta}_{OLS})$, and $\hat{\Sigma}_{OLS}$ as estimates of the coefficient matrix, the covariance matrix of the coefficient estimates, and the variance-covariance matrix of the reduced-form errors, respectively. Define the Cholesky factorization of $\hat{\Sigma}_{OLS}$ as $\hat{\Sigma}_{OLS} = ADA'$, where A is a lower unit triangular matrix and D is a diagonal matrix with the diagonal elements equal to the variances of the residuals (denoted as μ_0). Further, we stack the elements of A^{-1} below the diagonal row by row so that $\hat{a}_0 = [\hat{a}_{21,0}, \hat{a}_{31,0}, \hat{a}_{32,0}, \hat{a}_{41,0}, \hat{a}_{42,0}, \hat{a}_{43,0}]'$. We set the prior distributions of the model parameter as follows: (1) $\theta_0 \sim N[\hat{\theta}_{OLS}, 4 \hat{V}(\hat{\theta}_{OLS})]$, (2) $\ln(h_0) \sim N[\ln(\mu_0), 10 \times I_4]$, where I_4 is a 4 by 4 identity matrix, and (3) $a_0 \sim N[\hat{a}_0, \hat{V}(\hat{a}_0)]$, where $\hat{V}(\hat{a}_0)$ is a diagonal matrix and the each of the diagonal elements equals ten times the absolute value of the corresponding element. As noted by Benati and Mumtaz (2007) and Baumeister and Peersman (2013a, 2013b), the scaling of the variance-covariance matrices is arbitrary, but are used primarily to make the prior only weakly informative and to account for the relative magnitude of the elements in each matrix from the prior distribution. For hyperparameters, we

follow Baumeister and Peersman (2013a, 2013b) and assume that Q, S_i ($i = 1, 2, 3$) all follow an inverse-Wishart distribution such that $Q \sim IW(\bar{Q}^{-1}, T_0)$, $S_i \sim IW(\bar{S}_i^{-1}, i + 1)$, where T_0 equals the length of the training sample, $\bar{Q} = (0.01)^2 \hat{V}(\hat{\theta}_{OLS})$, and $\bar{S}_1, \bar{S}_2, \bar{S}_3$ are diagonal matrices with the relevant elements in $\hat{a}_0$ multiplied by 10^{-3} . Finally, the variances of the innovations in equation (7) are assumed to follow an inverse-Gamma distribution such that $\sigma_i^2 \sim IG(\frac{10^{-4}}{2}, \frac{1}{2})$, $i = 1, 2, 3, 4$.

Simulating the posterior distribution. The next step in implementing the Bayesian time-varying VAR model is to use the Markov Chain Monte Carlo (MCMC) algorithm to generate the joint posterior densities for four blocks of parameters: θ^T (coefficients), A^T (the contemporaneous correlation matrix), H^T (the variance), and M (the hyperparameters including the elements of $Q, S, \sigma_i^2, i = 1, 2, 3, 4$), where T represents whole sample up to period T . Specifically, posteriors for each block of the Gibbs sampler are generated conditional on the data Y^T and the other parameters obtained at previous steps. In step (1), we first obtain the joint posterior density of the VAR parameters using the Bayes rule conditional on A^T, H^T, M , and Y^T , and then compute their means and variances via forward and backward recursions. A similar algorithm is applied in step (2) to calculate the posterior of A^T conditional on θ^T, H^T, M , and Y^T . The volatility states ($h_{i,t}$) are drawn in the third step following the univariate algorithm of Jacquier et al. (1994). Given θ^T, A^T, H^T and Y^T , the conditional posterior distribution of the hyperparameters M can then be calculated. The joint posterior distribution $p(\theta^T, A^T, H^T, M | Y^T)$ is obtained by performing 100,000 iterations of the Gibbs sampler and discarding the first 50,000 draws.

Imposing sign restrictions. We obtain one random draw from the joint posterior of the time-varying lagged coefficients and hyperparameters from the Gibbs sampler at time t , and simulate the future paths of the four variables based on the variance-covariance matrix 20 quarters ahead. This accounts for all sources of uncertainty, including variations in lagged coefficients, changing fundamental relationships between the four endogenous variables, and additional shocks. To obtain $B_{0,t}$, we set $P_t D_t P_t'$ equal to the eigenvalue-eigenvector vector decomposition of the decomposition of the covariance matrix Ω_t in equation (8). A QR decomposition of K , an $N \times N$ matrix distributed $N \sim (0, 1)$, yields a matrix Q with columns orthogonal to each other. The structural impact matrix is then calculated as $B_{0,t} = P_t D_t^{1/2} Q'$, and together with a 4×1 vector ε_t distributed $N \sim (0, 1)$, the reduced form innovations are calculated $u_t = B_{0,t} \varepsilon_t$. The impulse response functions are calculated by finding the difference between the levels of the four variables with and without a shock. We retain only the impulse response functions that satisfy the entire set of sign restrictions, and iterate until we have a set of 500 estimates for each point in time. Median and the 16th and 84th percentile values are presented in our analyses. Section 5.3 adds additional

restrictions on permissible models based on elasticities estimated in Arora (2014), with only minimal differences in results.

4. Data

The data set adopted in this study follows Arora (2014) and Arora and Lieskovsky (2014) closely. Real natural gas prices are the US Producer Price Index (PPI) of natural gas published by the Bureau of Labor Statistics (BLS), in which October 1982 is used as the baseline. Compared to the conventional Henry Hub spot prices used in many previous studies, the natural gas PPI from BLS has the advantage of being a composite price of different natural gas prices in the US, and is available for a relatively long history that allows for a greater degree of freedom in econometric analysis. Additionally, it does not need to be deflated. To model the supply of natural gas, we use the marketed US natural gas production published by the EIA that measures the gas generated in the production process before liquids (such as propane and butane) are extracted. Arora (2014) argues that the marketed amount should be used rather than the gross amount to exclude the amount of gas consumed either in extraction or processing. Aggregate demand for natural gas is represented by an index of industrial production and capacity utilization (G.17) constructed by the Federal Reserve Bank. This demand is primarily driven by changes in US economic activity. Lastly, the EIA natural gas inventory data is used to measure the precautionary inventory demand in anticipation of changes in future supply-and-demand conditions.

The data period considered begins in 1976 and ends in 2015, and is sampled at a quarterly frequency. We use data from 1976Q1 to 1992Q4 as the training sample in the Bayesian time-varying VAR model, and use the data from 1993Q1 to 2015Q2 for time series analyses. Our choice of the time period used in separating training and estimation periods is motivated by the apparent structural change that occurred in the natural gas market in 1993. Under the Natural Gas Act of 1938, gas rates in the US had been traditionally regulated under the Federal Power Commission. Though the 1978 Natural Gas Policy Act attempted to deregulate the natural gas market, the Natural Gas Wellhead Decontrol Act of 1989 accelerated this process, formally ending all price regulations by January 1993. Combined with FERC order 636, which in 1992 mandated the unbundling of pipeline services (previously pipelines were able to combine sales and transportation), this effectively ended all government control over the market (Joskow, 2013). Finally, based on results from the Augmented Dickey-Fuller test, we transform the four variables into their logarithmic differences, which essentially measures the quarterly growth rate of each variable. To account for the apparent seasonality in consumption and inventory data, we further deseasonalize the four variables using seasonal dummies.

5. Estimation Results

In this section, we discuss the results from the Bayesian VAR model with time-varying parameters and stochastic volatility. Results are organized into sub-sections containing impulse response functions, elasticities, forecast error variance decompositions, historical decompositions, and analyses of historical events.⁷

5.1 Responses to Natural Gas Supply and Demand Shocks

Standard time-series analysis often reports the impulse response functions of endogenous variables given one standard deviation structural shocks. The resulting trajectory of impulse responses traces out the average reactions of current and future values of an endogenous variable to a shock to the current value of one of the structural innovations. Such an approach, however, may generate misleading results in a time-varying VAR model as not only the magnitude of each shock differs from period to period, but the propagation mechanism through which a structural shock affects the endogenous variables is not constant. It is impossible to know *ex ante* whether a change in response was due to a change in the relationship of the variable to each innovation or a change in the magnitude of an innovation. Here, we follow Baumeister and Peersman (2013b) and consider the less ambitious thought experiment of making the shocks comparable along some dimensions of observed market changes. We do not attempt to determine the sources of variations, but instead only to establish benchmark scenarios against which the changes in impulse responses can be compared across time periods. Specifically, in the first scenario, we normalize each structural shock such that each results in a 1% decrease in marketed US natural gas production. In the second scenario, impulse responses are normalized such that each structural shock generates a 10% increase in the real price of natural gas.

Figure 2 plots the median cumulative responses of four endogenous variables to each structural shock that lead to a 1% decrease in the marketed natural gas production in the US from 1994 to 2005, along with the 16th and 84th percentiles of their posterior distributions.⁸ Consistent with previous studies, results in figure 2 suggest that natural gas prices respond to shocks in supply and demand rather differently, and that the impact responses vary significantly over time. An unanticipated supply

⁷ A lag length of two is used in the estimation. We also considered longer lag lengths, but the results appeared to be qualitatively similar.

⁸ Note the responses of natural gas production are omitted from the plot because we have normalized its response to each structural shock to be -1%.

disruption, as shown in panel (a), generates a rise in natural gas prices. The magnitude of the impact response appears to be largest around 2000, increasing from 5% in the beginning of the sample, 1993, to its highest value of 12% in 2000. This dramatic increase in price responsiveness coincides with the formal market deregulation in 1993, possibly suggesting an adaptive learning process undertaken by market participants as they developed a better understanding of how market functions without government interventions. However, with the exceptions of Hurricane Katrina in 2005 and the 2007-2009 financial crisis, the impact response of unforeseen supply shocks has declined considerably after 2000. It is estimated that in 2015 a supply shock that leads to a 1% change in marketed natural gas production would result in approximately an 8% change in the real price of natural gas.

A negative supply shock is also associated with a reduction in aggregate real economic activity in the US and a decline in speculative inventory demand. However, the impact of supply disruptions on aggregate demand is rather limited overall. With few exceptions, the magnitude of response remained in a stable range between -0.4% to -0.2%. Even during the great recession of 2007-2009, the median impact response of aggregate economic activity to supply disruptions was at most -0.6%. Our results are consistent with a number of previous studies that find a limited role of natural gas supply shocks in domestic economic activities in the US (e.g., Kliesen, 2006). Results from time-varying impulse response functions also suggest that the impact of supply shortfalls on real economic activity may be exacerbated in a bearish market when the overall economy is shrinking. Though it is not possible to know the response of speculative inventory demand to an unanticipated supply shock *ex ante*, panel (a) of figure 2 suggests that during our sample period, inventory had been mainly drawn down to smooth out the imbalances between production and consumption caused by supply shortfalls. This negative impact responses apparently dominated the positive precautionary demand of inventory by forward-looking market participants who anticipate a potential gain for carrying natural gas into a future period.

Panel (b) of figure 2 displays responses to a negative shock in aggregate demand leading to a 1% decrease in marketed natural gas production. The shock resulted in a significant decrease in natural gas prices, ranging from 6% to 15% between 1994 and 2015. Consistent with results from supply shocks, a gradual decline in price responsiveness (smaller magnitudes) was observed after peaking in 2000, reflecting partly the ability of market participants to better manage price risks from unforeseen demand shocks in an increasingly maturing natural gas market after deregulation. Comparing panels (a) and (b) of figure 2, it is apparent that speculative demand responds differently to unanticipated shocks to supply and aggregate demand. A negative aggregate demand shock overall leads to an increase in speculative inventory demand, whereas a negative response is observed when the shock is originated from the supply side. In responding to unexpected disruptions in aggregate demand, storage holders build up inventory in

anticipation of tighter supply-and-demand relationships in the future. Our findings are consistent with previous theoretical work on storable commodities that inventory plays a vital role in mitigating price fluctuations due to supply and demand shocks (e.g., Working, 1949; Pindyck, 2001). However, it should be noted that starting from 2013, the increase in inventory after a negative aggregate demand shock is minimal, possibly reflecting market participants' bearish expectation that low natural gas prices are likely to prevail for an extended period of time with the apparent oversupply in the market.

Panel (c) of figure 2 suggests that an unanticipated negative speculative demand shock is needed to reduce marketed natural gas production. Following a decrease in speculative demand, aggregate demand associated with real economic activity declines marginally and the real price decreases. However, the magnitude of price changes from a speculative inventory demand shock appears to be significantly smaller than either a supply or an aggregate demand shock. Our results suggest that fundamentals remain the predominant determinants of the natural gas prices in the US, a result consistent with findings from the more extensively researched crude oil market (e.g., Kilian and Murphy, 2014). However, speculative inventory demand shock appears to play a slightly larger role after 2010. A similar magnitude of price response was also observed in the earlier part of the sample as the market underwent complete deregulation. Additionally, our findings suggest that positive speculative demand shocks have a similar effect on aggregate economic activity as unanticipated supply shortfalls.

In the second scenario, we normalize our impulse responses such that each structural shock raises the real natural gas price by 10%. Results plotted in figure 3 corroborates findings from figure 2.⁹ A negative supply shock resulting in a 10% increase in natural gas prices decreases the marketed production in the US between 0.5% and 2%, lowers the precautionary inventory demand by 2-3%, and reduces the aggregate economic demand by 0.2-0.4% during our sample period (panel (a)). The response of inventory demand appears to be of a slightly larger magnitude after 2007, contrasting with the impulse responses to a positive aggregate economic shock (panel (b), figure 3). As can be seen, inventory demand declines only slightly after a positive aggregate demand shock, and the magnitude of response has decreased in recent years. The disparities in impact responses highlight the critical importance of understanding the nature of structural shocks when analyzing natural gas prices. Though it generates the same responses in prices, speculative inventory demand responds rather differently to unforeseen structural shocks depending on where the shock originates: more inventory would be drawn into consumption to smooth

⁹ Note the responses of natural gas prices are omitted because we have normalized its response to each structural shock to be +10%.

out the imbalance between current supply and demand due to supply shortfalls, while aggregate demand shock virtually has no impact on inventory demand.

Additionally, we note in panel (b) of figure 3 that although the responses of natural gas production to a positive aggregate economic shock have changed over the sample period, their relative impacts in recent years are small. Due to the substantial capital investment required for natural gas production, a firm's ability to take advantage of a temporary positive aggregate demand shock is likely to be limited. However, given that our data are sampled quarterly, it is reasonable to assume that producers have some degree of flexibility to take advantage of market conditions. Nevertheless, our empirical estimates suggest that the magnitude of response may have been smaller than expected. Finally, we note that a positive speculative inventory demand shock is needed in order to generate an increase in prices. This positive inventory demand shock would also lead to an increase in production and a decline in aggregate demand. These impact responses have both been relatively stable in recent years.

5.2. Estimating Supply and Demand Elasticities in the US Natural Gas Market

Direct estimation of price elasticities in the natural gas market is difficult as the observed movements in quantity and price are often driven by a combination of supply and demand factors that are hard to disentangle. However, in a structural VAR model, structural shocks are by definition mutually uncorrelated, each generating unique reactions in the endogenous variables, with each having direct economic interpretations. In our model, we have identified structural shocks by imposing unique sign patterns in the contemporaneous responses of endogenous variables. Since these structural shocks induce responses in the natural gas market variables by shifting either the demand or supply curve, we can calculate the implied short-run price elasticities from impulse responses associated with each structural shock by dividing the percentage changes in quantity with the percentage changes in prices. Figure 4 presents the evolution of median short-run price elasticities of natural gas supply and demand in the US between 1994 and 2015, along with their 16th and 84th percentile posterior distributions. Note that in our analysis we consider three different dimensions of demand shocks each of which can shift the demand curve upward or downward. Hence we trace out the curvature of the supply curve associated with aggregate demand and inventory demand shocks. As can be seen, elasticity estimates presented in figure 4 confirm our findings from impulse responses that the supply of natural gas in the US responds differently depending on the nature of the demand shock. Except the early part of the sample, the median supply elasticities associated with aggregate demand shocks have been rather stable, oscillating between 0.06 and 0.1 (panel (a), figure 4). Even during the period of rapid shale production, the supply elasticity associated with aggregate demand shock has only increased slightly, settling at around 0.1 in 2015.

The evolution of supply elasticity due to speculative demand shocks is presented in panel (b) of figure 4. Much larger variations are observed for the first ten years of the sample (1994-2003), with the median supply elasticity ranging from 0.15 to 0.3. Since then, the curvature of the supply curve derived from speculative demand shocks has remained somewhat stable, fluctuating around 0.25, with small increases observed after 2010. Results suggest that natural gas supply in the US may have been more responsive to precautionary inventory demand shocks than aggregate demand shocks. Additionally, though natural gas supply in the US is inelastic overall, elasticity has improved slightly since the shale boom. Arora (2014) argues that such increases in elasticity should not be surprising as production can take place faster when extracting natural gas from shale formation as compared to conventional production methods. Overall the increase in supply elasticity is small.

Panel (c) of figure 4 plots the median price elasticities of demand derived from shifts in the supply curve.¹⁰ As can be seen, the demand elasticity ranges from -0.08 to -0.18 over the sample period. After a decrease in the magnitude of elasticity (steeper demand curve) in the first five years of the sample, the demand elasticity in the US natural gas market remained relatively stable for the following 15 years, fluctuating around -0.10. Recently, however, the demand curve has somewhat flattened, possibly reflecting a greater flexibility in energy substitution as industrial, commercial, and residential users are better able to switch among different fuel sources. As a result, a similar price change induces a larger impact in quantity demanded than before.

Our elasticity estimates are in general consistent with the numbers reported in Arora (2014) using quarterly data. Arora (2014) finds that over the period of 1993-2013, the median short-run demand elasticity estimate is -0.14, and that the median short-run supply elasticities estimated with aggregate demand and speculative demand shocks are 0.05 and 0.13, respectively. Excluding the shale boom period (1993-2007), Arora (2014) finds that while quarterly demand elasticities remained similar, the supply was considerably less elastic (0.01 and 0.10 for elasticities derived from aggregate demand and inventory demand shocks, respectively). Our results, by contrast, suggest a smaller increase in price elasticity of supply after the shale production boom. Additionally, our results are comparable with several previous supply elasticity estimates in the US natural gas market. Dahl (1992) and Barret (1992) estimated domestic price elasticity of supply to equal 0.41 and 0.014 respectively, however both studies use data sampled at the yearly interval. Krichene (2002) estimated the long-run price elasticity of supply at the world level to be 0.6, and at the short-run elasticity to be -0.06 during 1918-1999. At the monthly interval, Ponce and Neumann (2014) estimate that after one month, the price elasticity of supply is equal to 0.056, similar to our estimate of response associated with aggregate demand.

¹⁰ Kilian and Murphy (2014) and Arora (2014) differentiate price elasticity of demand between those in use and in production. Here we only consider the price elasticity of demand in production

5.3 Robustness Checks

Kilian and Murphy (2012, 2014) argue that using sign restrictions alone only provides weak identification of structural shocks in a VAR model. In addition to sign restriction, elasticity bounds should be used to reduce the set of admissible models. By retaining only models that produce plausible estimates of elasticities, it can be ensured that results are associated with economically reasonable interpretations. However, as identified in Arora (2014), very few papers in the literature have investigated the quarterly supply elasticities of the natural gas market in the US, and the demand elasticity has mostly focused on the residential demand side. For a check on the robustness of our results, we use twice the median elasticity estimates of Arora (2014) obtained for the 1993-2013 period as a bound to construct ranges for model admission. Specifically, we postulate that the short-run elasticity of demand elasticity cannot exceed -0.28, and that the supply elasticities associated with aggregate and speculative demand shocks cannot exceed 0.1 and 0.26, respectively.

Figure 5 plots the median impulse response functions from admissible models that fall within the elasticity bounds, normalized on 1% decreases in marketed natural gas production. In general, results appear qualitatively similar when no restrictions are imposed. A comparison between figures 2 and 5 suggests that shocks to production are virtually identical with or without bounds. While there are noticeable differences in results with other impulse response functions, they maintain the movements, and are only a small percent difference. Two exceptions are the price responses to aggregate and speculative demand shocks (last chart of panels (b) and (c)). Larger magnitudes are observed for these two impulse responses toward the end of the sample, indicating that aggregate demand and speculative inventory shocks are playing a greater role in recent years. These results are not unexpected as rather stringent restrictions are imposed on demand and supply elasticities—a similar shock would on average generate larger price responses. Regardless, our results without elasticity bounds are qualitatively comparable to those with elasticity bounds.

6. What Drives Variations in the Real Price of Natural Gas?

Impulse response functions and elasticity estimates provide useful information on the responses of endogenous variables following a one-time structural shock. However, they do not consider the relative importance of each structural shock on the evolution of price movements over time. To shed light on this issue, we investigate the forecast error variance decompositions (FEVD) and historical decompositions (HD) of natural gas prices following the structural VAR estimation.

Figure 6 plots the median contribution of each structural shock to the forecast error variance of real prices, along with their 16th and 84th percentile posterior distributions. It can be seen that the explanatory power of supply shocks peaked in 1999, 2005, and again in 2013, accounting for approximately 35 percent of the error variance of real natural gas price forecasts. Noticeable declines are observed during the great recession and another after 2013. Regardless, the relative contribution of supply disruptions to the total variance of the natural gas price forecast error has been consistently above 25% throughout the sample period, suggesting the important role played by supply shocks in the US natural gas market since deregulation.

Panel (b) of figure 6 suggests that a similar fraction of the price forecast error variance is attributable to shocks in aggregate economic demand prior to 2009, after which its explanatory power dramatically declined to about 20%, though a small rebound was observed at the very end of the sample. By contrast, with the exception of the pre-2000 period, residual shocks (other natural gas market demand shocks not accounted by the other three shocks) have consistently accounted for about 30 percent variation in the natural gas price movements. Results suggest that much of the natural gas price variability in the US may be due to market-specific demand factors such as consumers' preference changes, technology improvement that enabled more efficient uses of household appliances, substitution with other energy sources in electricity generation, demographic movements, etc.

Compared with the other three shocks, the relative importance of precautionary inventory demand shocks in shaping price movements has changed considerably. As shown in figure 6, close to 20% of the forecast error variances of prices can be explained by inventory demand shocks in mid-1990s. For the next decade, however, only between 5-8% of the error variance is attributable to inventory demand shocks. In recent years its explanatory power has increased again, accounting for over 10% of the variation in natural gas prices in 2015. Our results highlight the important role that inventories perform, regulating the prices of storable commodities as forward-looking agents increase or decrease inventory in anticipation of changes in future supply-and-demand conditions. However, it should be noted that precautionary inventory demand shocks remain a small force in the US natural gas market compared to shocks from fundamentals.

Figure 7 plots the median historical contribution of the four structural shocks to the evolution of natural gas price returns in the US. The solid black line shows the deviation of the deseasonalized natural gas price returns from its mean, and the dashed blue line traces the cumulative effect of the specific structural shock to the evolution of natural gas prices assuming occurrence of no other structural shocks. In other words, the historical decomposition indicates how the price of natural gas would have evolved

had only one structural shock occurred.¹¹ As is evident in figure 7, the cumulative effect of each structural shock to natural gas price fluctuations has changed significantly from period to period. Below, we examine three historical episodes characterized by large price volatility to shed light on the relative importance of each structural shock in the US natural gas market.

6.1 How Much Did Shale Production Contribute to the Collapse of Natural Gas Prices?

Technological advances in drilling methods have made it profitable to extract natural gas from shale and other low-permeability formations. Since 2006, the popularization of combined hydraulic fracturing and horizontal drilling technologies have revolutionized the US natural gas market. Prior to the mid-2000s, natural gas production in the US had been on a slow decline for more than three decades. However, between 2006 and 2015, US dry natural gas production has increased by 39.04%¹², largely attributable to the substantial increase in gas from unconventional sources. Meanwhile, natural gas prices in the US have plummeted, dropping more than 82% from their peak in 2008 at approximately \$10.79 per thousand cubic feet to \$1.89 in May 2012¹³, though a small rebound was observed afterwards. Economists and market analysts have largely attributed such dramatic price declines to expansion in shale gas production.

Our results in figure 7 indicate that shocks to natural gas production have played a much more significant role in recent years compared to the pre-shale era. Their importance is particularly evident after 2010, when the line indicating projected price movements with only supply shocks virtually overlaps the actual price line during this period. However, the role of aggregate demand and speculative inventory demand shocks cannot be ignored. As is evident, price movements assuming only aggregate demand shocks have largely followed the same trend as the actual price fluctuations in the post-shale period, highlighting again the importance of real economic activities in the natural gas market. Further, it appears that precautionary inventory demand shocks could explain part of the price movements at the end of 2012, and again in 2014.

6.2. The Price Escalation in 2007-2008

Among all the historical events in the US natural gas market that experienced a dramatic price volatility, its price escalation in 2007-2008 is perhaps the least understandable and the most complex

¹¹ Note that since we deseasonalized our data prior to estimating the time-varying VAR model, the solid black line may behave differently from the actual price movements.

¹² See EIA dry production data at: <https://www.eia.gov/dnav/ng/hist/n9070us2a.htm>.

¹³ See EIA natural gas price data at: <https://www.eia.gov/dnav/ng/hist/n9190us3m.htm>.

(e.g., Smead, 2010). In the beginning of 2007, wellhead prices in the US were still fluctuating between \$6 and \$7 per thousand cubic feet. However, by the summer of 2008, the price of natural gas had almost doubled, exceeding \$10 per thousand cubic feet¹⁴. At the same time, shale production started to expand and the market was in fact in over-supply. EIA data suggested that the domestic gas production in the first quarter of 2008 was almost 10% higher than the same quarter of 2007, with consumption rates (the difference between total availability and inventory) essentially flat. This appears to be inconsistent with classic demand and supply theory. So, why did the price of natural gas increase so much in such a short time period?

To understand this phenomena, three relevant facts need to be considered. First, historically the price of natural gas has been closely linked to oil prices. The two prices have been linked for more than two decades since the formal deregulation of the natural gas market in 1993. Economists have generally believed that there exists a long-run relationship (cointegration) between the two price series. Crude oil prices started their run in 2007, increasing more than 245% in less than two years, peaking at \$133.88/barrel in the summer of 2008 before plummeting to \$41.12/barrel in December 2008¹⁵. Second, a large number of commodities, including energies, grains, softs, and metals all experienced similar price run-ups during this period. A common view in the literature is that speculators partly, if not fully, caused the systematic price increases in commodity markets. Third, even prior to the financial crisis, the housing market had already started its collapse in 2007. Previous literature has proposed a “bubble migration” theory in which financial bubbles first migrated from real estate to the bond market, and then subsequently to the commodity markets (e.g., Phillips and Yu, 2011). In other words, the price rises and falls may have been linked to the overall business cycles in the US and in the world.

Figure 7 suggests that speculation may have indeed *partly* affected natural gas price movements in 2008. Assuming only speculative inventory demand shocks, the trajectory of price movements (blue line) appears to closely track the actual price behavior (black line). A similar result may be found in figure 6, which indicates a small increase in the explanatory power of inventory demand shocks on price forecast error variance during this period. In anticipating tighter future demand-and-supply conditions, market participants may have increased their inventory to be carried into future periods for potential profits. However, such an impact is only short-lived, as the actual price behavior drifts away from the path projected by only inventory demand shocks. By contrast, supply and aggregate demand shocks account for the bulk of the price volatility during this period. The projected price movements assuming

¹⁴ See EIA natural gas wellhead price data at <https://www.eia.gov/dnav/ng/hist/n9190us3m.htm>.

¹⁵ See EIA crude oil data at <https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=RWTC&f=A>.

only supply or aggregate demand shocks appear to line up closely with the line indicating actual price movements.

Overall our results are consistent with previous studies investigating the speculative influences in the natural gas market. Bohl and Stephan (2013) find that financialization in the natural gas market did not significantly increase spot price volatility, while Manera et al. (2014) reported that in the short-run speculation did partially increase price volatility, though this effect becomes negative in the long-run. Meanwhile Geman and Ohana (2009) stated that the correlation between spot price volatility and natural gas inventories is negative only in periods of scarcity, when inventory is below its long run average for natural gas. Results from this study showed that though the impact of speculation cannot be ignored, it plays a very limited role in natural gas price fluctuation in 2008. Instead, price movements predominantly reflect changes in fundamentals.

6.3. Hurricanes Katrina and Rita in 2005

Significant supply disruptions occurred to the US natural gas market in the second half of 2005 when hurricanes Katrina and Rita hit the Gulf of Mexico. Before the hurricanes hit, Louisiana and the federal offshore production areas in the Gulf accounted for approximately 20% of the total natural gas production in the US¹⁶. These back-to-back hurricanes damaged offshore platforms, major processing facilities, and major pipeline segments, resulting in the loss of over 6 billion cubic a feet per day of offshore natural gas supply for an extended period of time. Even months after the storms, the equivalent of 10% of US consumption remained shut-in due to problems with transportation and production (Kumins and Bamberger, 2005). These prolonged supply disruptions put extreme pressure on the supply-demand conditions in the US natural gas market, as fears rose in anticipation of heightened winter heating costs. As a result, the price of natural gas exceed \$10.3 per thousand cubic feet in October 2005. However, prices dropped quickly, reaching \$6.85 in February 2006 as winter that year turned out to be much warmer than anticipated. Low natural gas prices lasted into the rest of the 2006 until the spike in 2008.

This explanation implies that supply shocks in the structural model should account for most of the US natural gas market price surge in 2005. Results in figure 6 suggest that the implied price movements assuming only supply shocks in 2005 line up closely with the actual path of price behavior. Precautionary inventory and aggregate demand shocks, on the other hand, do little to explain the initial price run-ups in

¹⁶ See EIA Gulf of Mexico dry natural gas production data at: http://www.eia.gov/dnav/ng/hist/rngr20r3fm_1a.htm, and Louisiana dry production data at: <http://www.eia.gov/dnav/ng/hist/n9010la2m.htm>.

2005. The story changes in 2006, when it appears that aggregate demand shocks do partly explain the natural gas price plunge.

7. Conclusions

An integral component of the US economy, the natural gas industry has undergone considerable changes over the past two decades, with the price of natural gas becoming increasingly volatile over time. Effective policy responses to these price changes and other exogenous shocks require a thorough understanding of the underlying drivers behind natural gas price movements. The purpose of this paper is to shed light on this issue by analyzing the time-varying effects of supply and demand shocks on the natural gas market in the US from 1993 to 2015 using a structural vector autoregression model. Specifically, we disentangle the real natural gas prices into four structural shocks: unanticipated supply shocks, aggregate demand shocks, precautionary inventory demand shocks, and residual shocks. Previous studies using a similar approach suffer from four apparent drawbacks: (1) ignoring the time variation in the model parameters, (2) assuming discrete abrupt structural breaks, (3) assuming residual homoscedasticity, and (4) imposing a recursive causal chain in contemporaneous correlations. In this paper, we follow Baumeister and Peersman (2013a, 2013b) and seek to address these four problems using a sign-identified structural vector autoregression model that allows for both drifting parameters and time-varying volatility.

Overall, we find that natural gas prices respond to shocks in supply and demand rather differently, and the impact responses differ rather significantly over time. An unanticipated supply disruption raises natural gas prices, reduces the aggregate economic demand, and decreases the precautionary inventory demand. Its impact on prices peaks around 2000, but has decreased since then with a few exceptions. Though supply shortfalls overall have a limited impact on aggregate economic demand, their negative effect may have been exacerbated in a bearish market when the overall economy was shrinking. Inventory demand decreases in response to supply disruptions, apparently dominating the positive precautionary demand of inventory by forward-looking market participants who anticipate a potential gain for carrying natural gas into a future period.

A negative aggregate demand shock, on the other hand, depresses natural gas prices, reduces natural gas production, and encourages precautionary inventory demand. Consistent with findings from supply shocks, a gradual decline in price responsiveness (smaller magnitudes) was observed after peaking in 2000. The response of inventory demand to aggregate demand shock is relatively stable between 2000 and 2012, after which it has significantly declined. The limited response of inventory demand since 2013

possibly reflects the bearish view of market participants toward natural gas prices in anticipation of a prolonged period with abundant natural gas supply. We also find that following a negative precautionary inventory demand shock, aggregate demand associated with real economic activity declines marginally, and both marketed natural gas production as well as real prices also decrease. However, the magnitude of price responses from a speculative inventory demand shock appears to be significantly smaller compared to either a supply or an aggregate demand shock.

Based on the estimated impulse response functions, we calculate the price elasticities of supply and demand. With the exception of the early part of the sample, the median supply elasticity associated with aggregate economic demand shocks is estimated to be between 0.06-0.1. The median supply elasticity associated with speculative inventory demand shocks appear to be of larger magnitudes, ranging from 0.15 to 0.3. For both supply elasticities a small increase in magnitude is observed toward the end of the sample, possibly due to the fact that new drilling technology has enabled increased production. The price elasticities of demand are estimated to be between -0.18 and -0.08 over the sample period. Similarly, the magnitude of demand elasticity increased in recent years, possibly indicating that in the new era characterized with ample natural gas supply, industrial and residential users may have greater flexibility of substitution in fuel use. Our elasticity estimates overall appear to be consistent with the numbers reported in Arora (2014) using quarterly data.

We further investigate the relative importance of the four structural shocks in driving natural gas price movements using forecast error variance and historical decompositions. We find that supply shocks consistently account for over 25% of the price variation in the post-regulation era, with its role peaking in 1999, 2005, and again in 2013. The explanatory power of aggregate demand shocks is high as well, typically accounting for over 20% of the price forecast error variance over time. The fraction of forecast variance attributable to speculative inventory demand appears to be the highest in 1994, followed by a significant decrease in the next 15 years, though a gradual increase is observed after the shale boom. Nevertheless, the explanatory power of precautionary inventory demand shocks is small compared to the other shocks. We also note that there is a large portion of forecast error variance that cannot be explained by supply, aggregate demand, and speculative demand shocks.

To put our impact estimates of supply and demand shocks into perspective, we examine three historical episodes. The recent shale boom constitutes the first episode, during which the production of natural gas skyrocketed and its price plummeted due to improved drilling technologies. We find that shocks to natural gas production can explain the bulk of the price movement in this period, a finding consistent with the general view of market analysts. However, the role of aggregate demand and

speculative inventory demand shocks cannot be overlooked, as these two shocks combined could explain part of the price movements at the end of 2012 and in 2014.

In the second historical episode, we investigate the 2007-2008 price escalation in the US natural gas market when the production was in fact larger than usual. We find that speculation can at least *partly* explain the price variation during this period. However, contrary to common belief, the impact is only short-lived and of small magnitude. By contrast, supply and aggregate demand shocks can account for the bulk of the price volatility during this period. In the third episode, we investigate the 2005 price spike during Hurricanes Katrina and Rita, and find that the implied price movements assuming only supply shocks in 2005 line up closely with the actual path of price behavior.

Our analyses provide a comprehensive and innovative framework for understanding the demand and supply shocks and their impacts on prices in the US natural gas market. Results in this study highlight the importance of incorporating time variations in model parameters. Not only have the impulse responses evolved over time, the relative importance of structural shocks in driving natural gas price movements vary rather significantly in different periods as well. Such variability in model parameters, if ignored, could potentially misinform policymakers and lead to policies detrimental not only to the natural gas industry but also the economy as a whole. The estimation and model identification strategy outlined in this study enables us to address this issue by more accurately gauging the effects of supply and demand shocks in the natural gas market over time.

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Tables and Figures

Table 1. Sign Restrictions Imposed in the Structural VAR Model

	Supply shock	Aggregate demand shock	Speculative inventory demand shock
Marketed production	+	+	+
Aggregate demand	+	+	-
Inventory			+
Real price	-	+	-

Note: All structural shocks are assumed to be positive. Missing entries imply that no sign restriction is implied in the corresponding impulse response function in the immediate period.

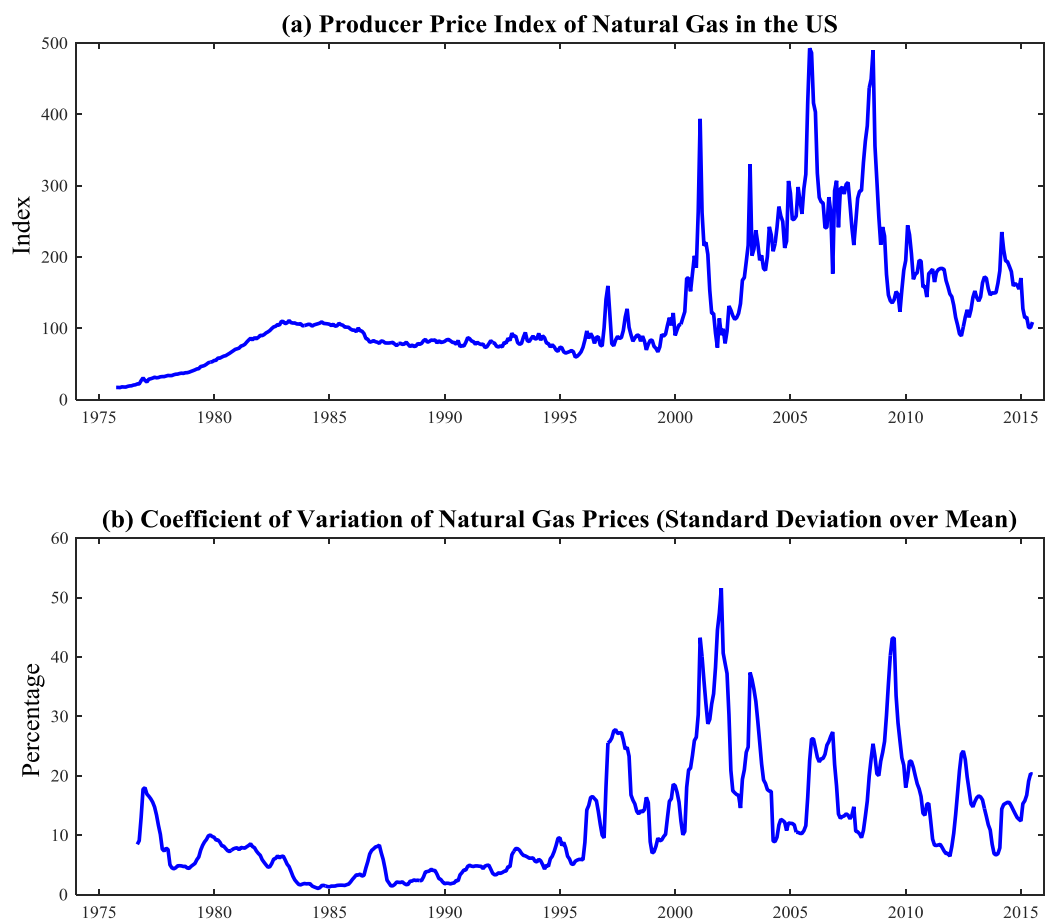


Figure 1. Natural Gas Producer Price Index in the US and Its One-Year Coefficient of Variations (1975-2015)

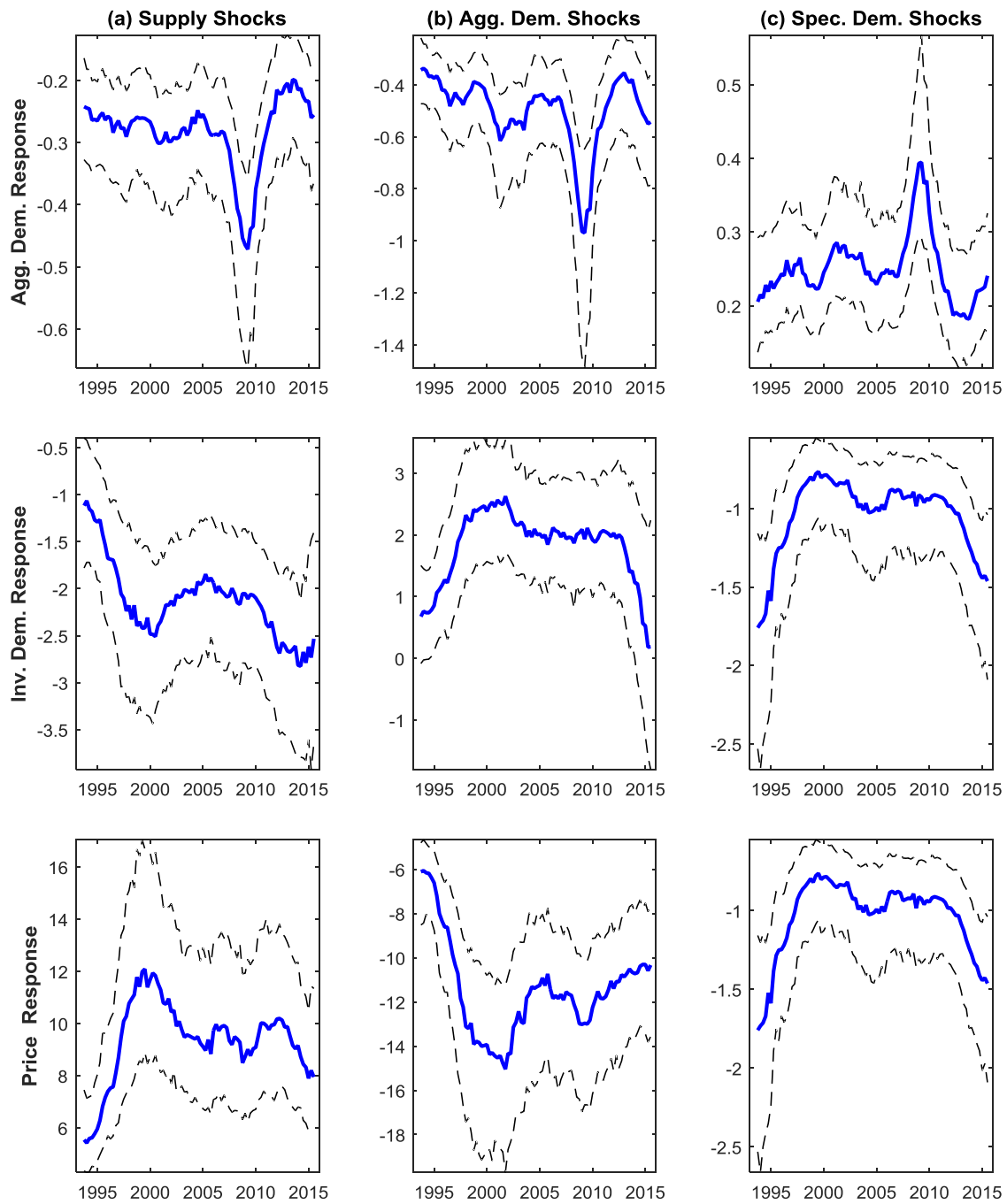


Figure 2. Median Cumulative Impulse Responses of Aggregate Demand, Inventory Demand, and Price to Structural Shocks Normalized on 1% Supply Shortfalls

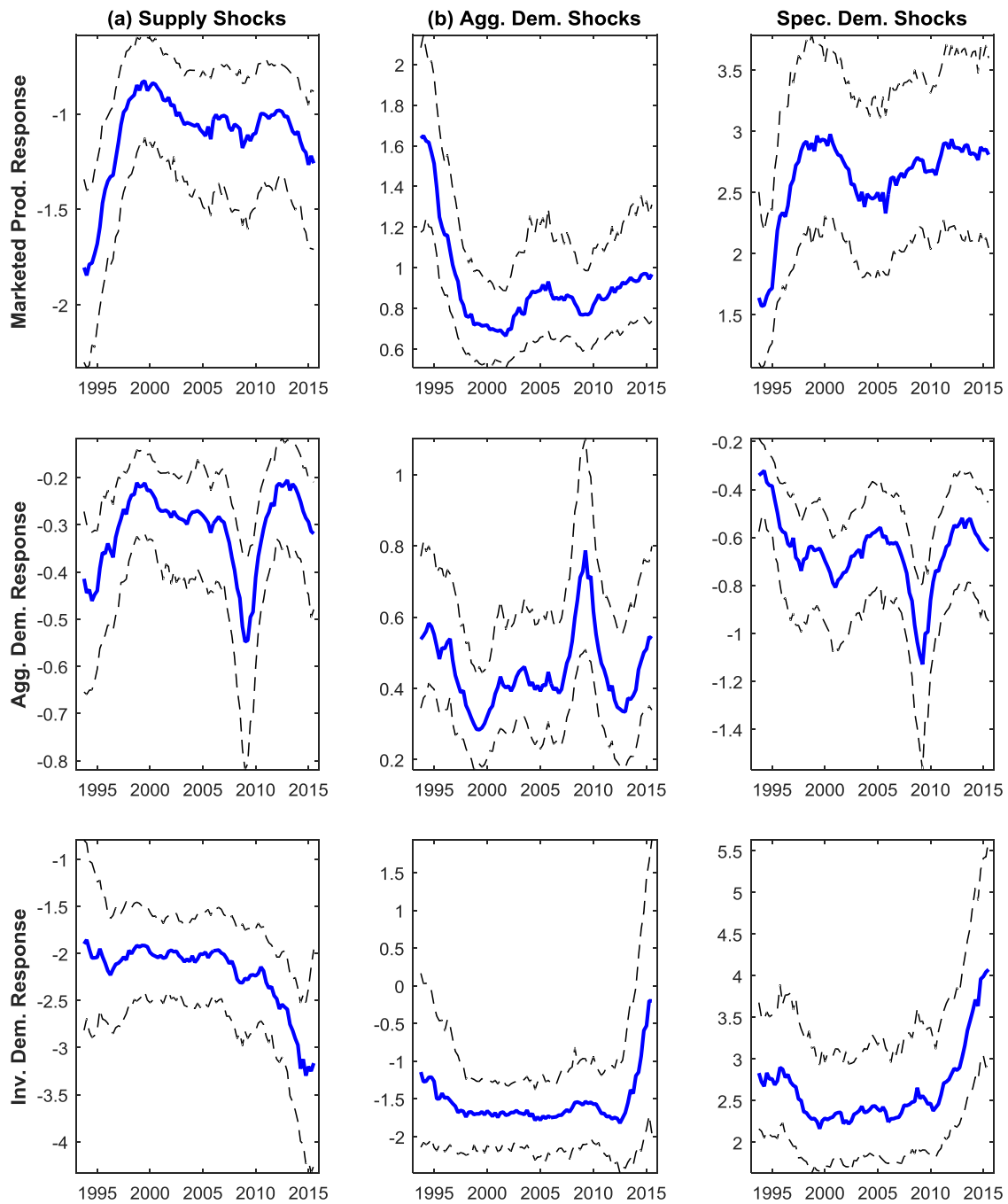


Figure 3. Median Impulse Responses of Aggregate Demand, Inventory Demand, and Price to Structural Shocks Normalized on 10% Price Increase

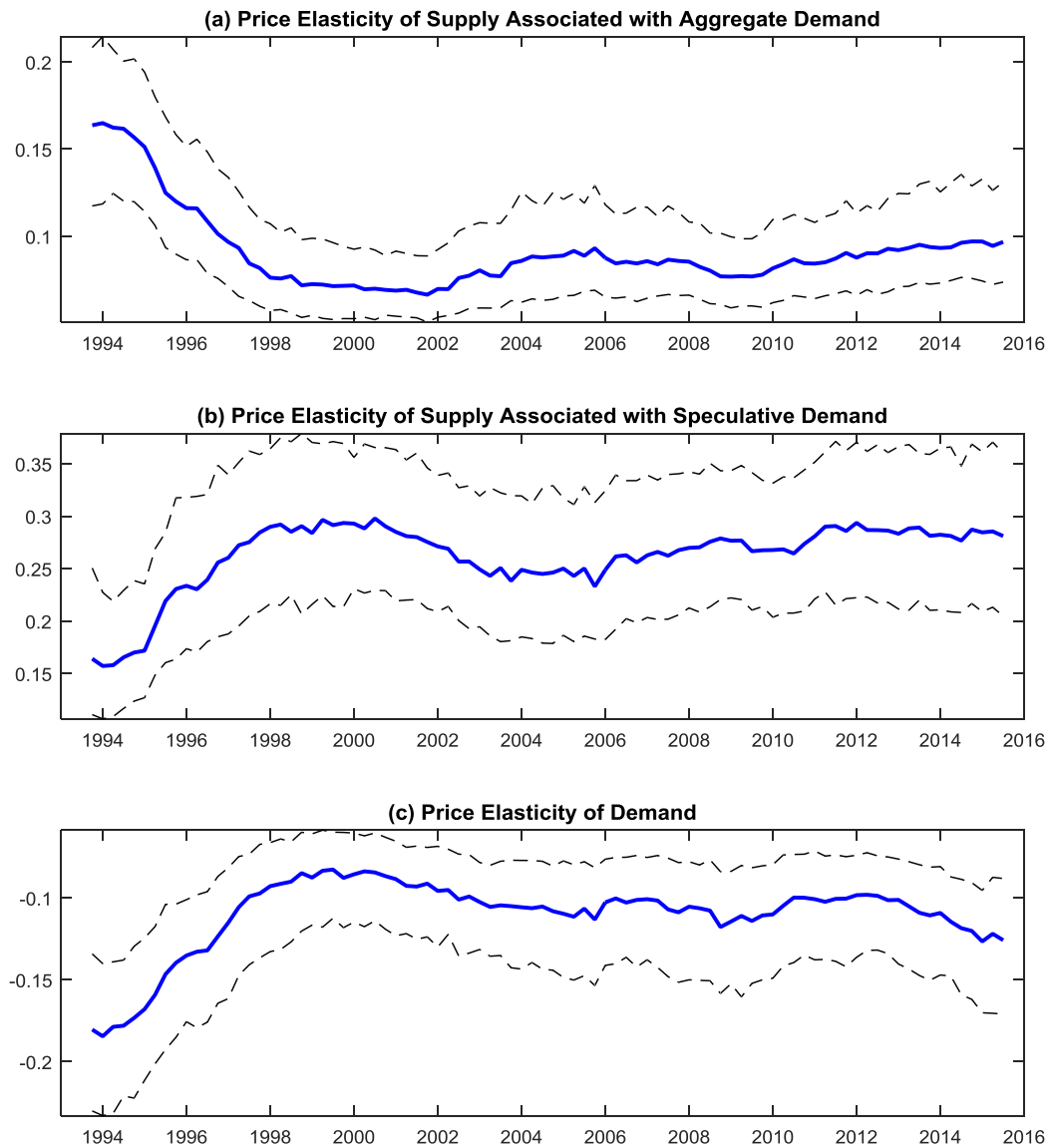


Figure 4. Median Natural Gas Price Elasticities of Demand and Supply Derived from Impulse Responses Estimates (1993-2015)

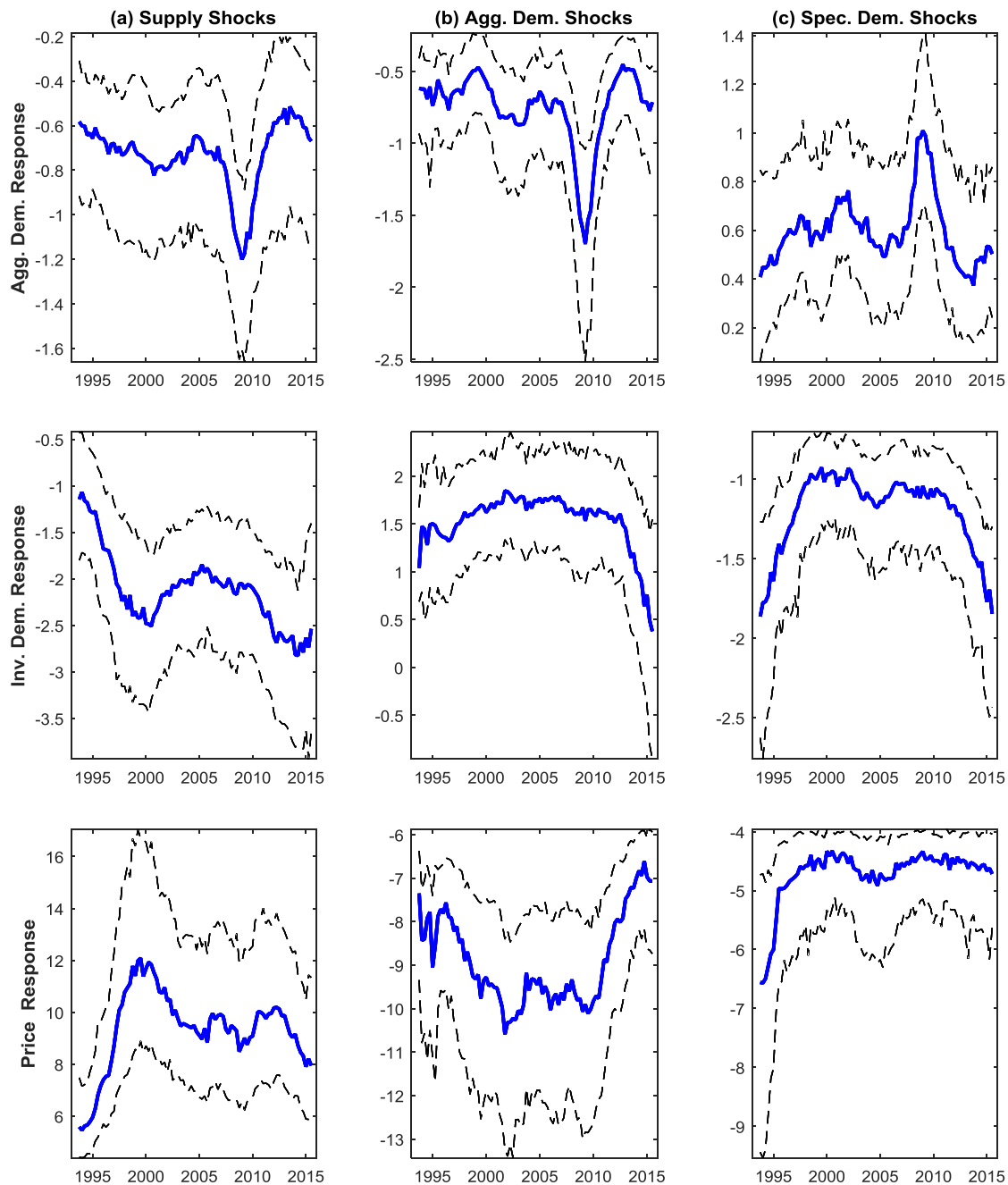


Figure 5. Robustness Check – Median Cumulative Impulse Responses of Aggregate Demand, Inventory Demand, and Price to Structural Shocks Normalized on 1% Supply Shortfalls after Imposing Elasticity Bounds

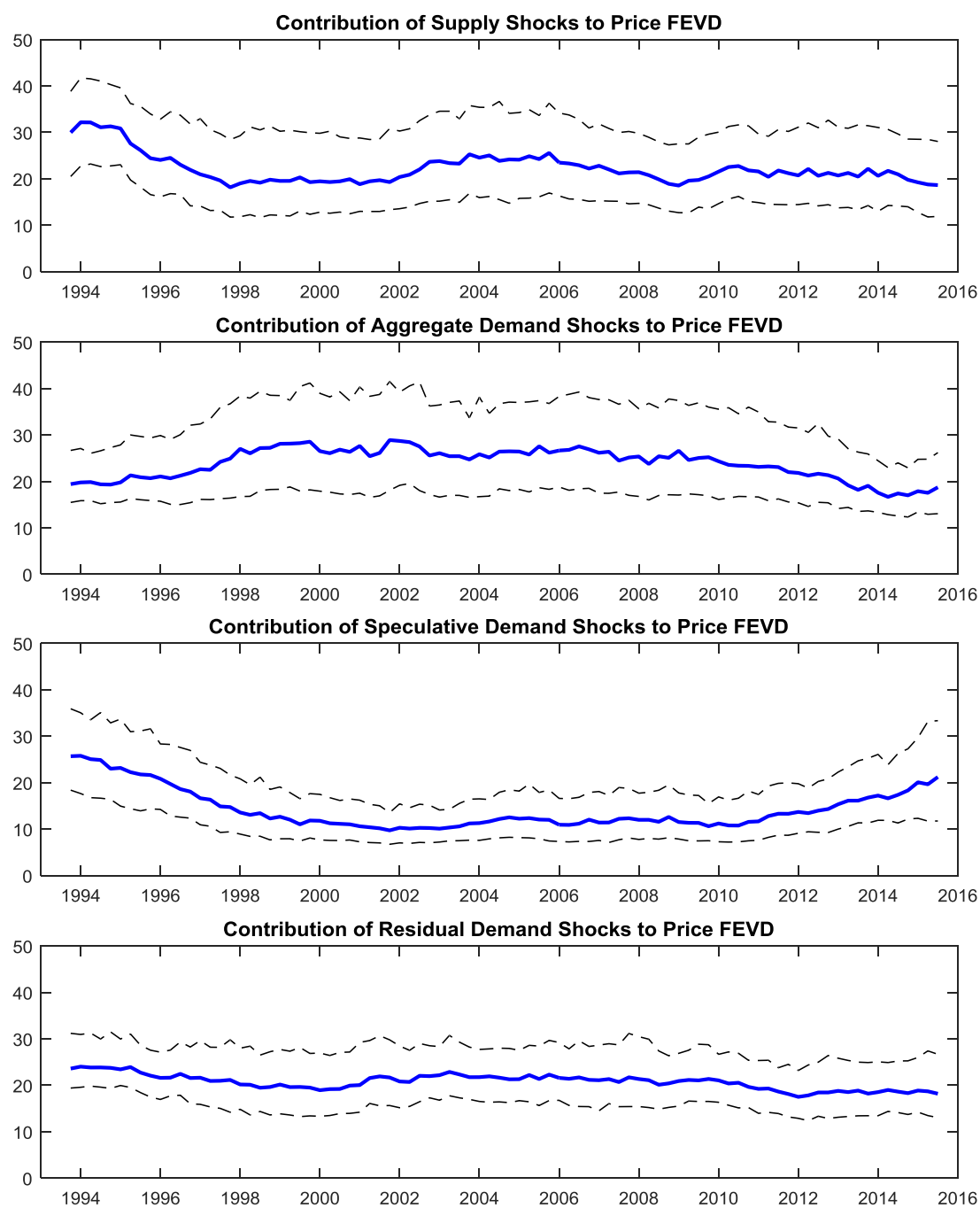


Figure 6. Medium Forecast Error Variance Decomposition of Real Natural Gas Prices in the US (1993-2015)

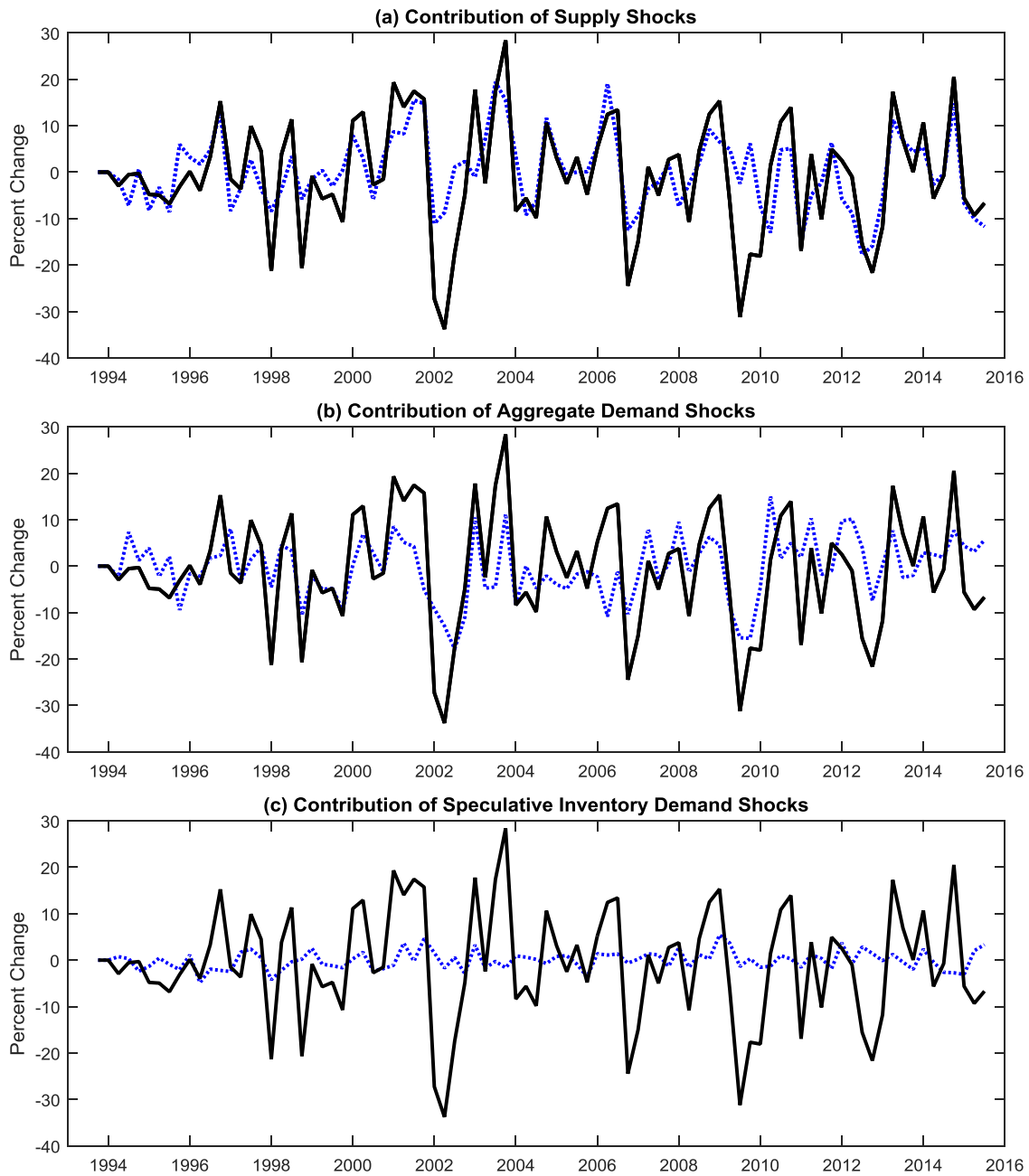


Figure 7. Historical Decomposition of Demeaned Real Natural Gas Price Changes (Demeaned Data in Black and Contribution of the Structural Shock in Blue)